Jeremyah Joel

Security Engineer

joeljeremyah@gmail.com | LinkedIn | Technical writing | VigilX

Security engineer across banking, fintech and national education platforms. Built cloud detection and internal red-team capabilities from scratch. Hands-on experience in application security, cloud monitoring, anti-fraud engineering and security automation, including detection and incident response support.

Certifications

OffSec Web Expert (OSWE) | OffSec Certified Professional (OSCP)

CREST Registered Penetration Tester (CRT) | CREST Practitioner Security Analyst (CPSA)

Selected Results

- **Cloud detection engineering** Built detection across a legacy GCP estate with limited visibility and budget; the team used it to identify and remediate high-risk vulnerabilities.
- **Google Cloud Armor** Discovered and reported a WAF bypass affecting Classic Application Load Balancers; Google subsequently released a global patch.
- **15+ reported security findings** over six months in heavily audited open-source protocol and validator codebases, including high-impact vulnerabilities.
- **Credential exposure intelligence** Built SHADOW to query 4.19 billion canonical credential records on a 16 GB Mac Mini; measured 0.38 seconds for one warm, single-threaded domain lookup.

Professional Experience

DKATALIS | Staff Engineer, Application Security Aug 2024 - Present

- Own offensive security and assessment automation for a digital bank serving 15M+ users; report directly to the CISO.
- Built the internal red-team capability from scratch; conduct reverse engineering, exploit development and anti-fraud engineering on financial infrastructure.
- Work across penetration testing, DevSecOps and SecOps: embed security into CI/CD and support detection and incident response.
- Introduced AI-assisted assessment automation, reducing repetitive assessment work by approximately 70%.

GovTech Edu | Senior Product Security Engineer Oct 2022 - Aug 2024

- Led product security across five national education programs serving 53M+ users, including threat modeling, architecture review and pre-launch assessments.
- Built detection engineering across a legacy cloud estate with limited visibility and budget, covering access-control abuse, injection and lateral-movement threats.
- Combined Google Cloud request and database logs, Security Command Center, Ops Agent and Linux auditd; tested detection scenarios in a sandbox.
- Strengthened application-level monitoring to detect database-error patterns and investigate gaps in cloud security controls.

Security Research and Engineering

VigilX (Side project) | Founding Engineer Jan 2026 - Present

- Built an agent-assisted audit harness combining code mapping, explicit threat models, context selection, model routing, verification gates and persistent investigation records.
- Implemented persistent state, idempotent processing and retries to retain evidence, failure causes and unfinished investigations for re-examination after execution failures.
- Conduct low-level C and protocol research, including Firedancer; use tests, sanitizers, crash reproduction, fuzzers and invariant checks where applicable, with human review before reporting.

Earlier Experience

GoTo Group | Product Security Engineer 2022

- Led offensive security for post-merger integration across financial platforms; built real-time asset monitoring across the merged API and cloud estate.
- Integrated open-source security tooling into CI/CD and worked with governance teams on PCI DSS controls for payment services.

Vantage Point Security | Security Consultant 2021 - 2022

- Delivered security assessments, client-facing reports and briefings for 10+ banking and financial institutions across web, API, mobile and hardware, including kiosks and ATMs.
- Found critical vulnerabilities in Indonesian regulatory infrastructure and presented findings to client stakeholders and leadership.

Education

Master of Business Administration, Quantic School of Business and Technology

Bachelor of Computer Science, Cyber Security, BINUS University

Selected Publications

Strategic Detection Engineering | INA Digital Edu, 2023
Cloud detection using application, database and operating-system telemetry.

Google Cloud Armor Vulnerability Discovery | INA Digital Edu, 2024
Investigation, provider escalation and monitoring improvements for a WAF bypass.

Precompute, Don't Search: 4.19 Billion Leaked Credentials in 0.38 Seconds | July 2026
Memory-bounded credential intelligence, deduplication and exact-domain indexes at billion-record scale.

The harness is the economics | October 2026
Research verification, recoverable execution and capacity allocation in an AI-assisted audit pipeline.

LLMs Are Non-Deterministic. Your Security Pipeline Shouldn't Be. | GoPenAI, March 2026
Scoped code context, model orchestration, retries and structured-output handling for security analysis.